

XIE Jintao

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PROFILE

Year 4 Data Science & Engineering student at HKU with a Minor in Digital Health. Bridges AI engineering with healthcare domain knowledge — building ML systems for clinical data, mental health applications, and medical research. Comfortable across the full stack: from PyTorch model training to production deployment. Seeking to apply AI to real-world patient care at Rapport.

TECHNICAL SKILLS

AI & ML PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, LangChain, computer vision, NLP, statistical modeling
Languages Python (primary), R, SQL, TypeScript
Data & Infra pandas, numpy, DuckDB, ChromaDB, SQLite, Flask, API integration, time series analysis
Healthcare Digital Health coursework, EEG/clinical data analysis, health informatics, data privacy
Tools Git, Chrome Extensions, Next.js, Linux, Arduino, ESP32

PROJECTS

EEG-Based Computational Psychiatry & Accessibility

Jan 2026 – Present

HKU Innovation Wing, Faculty of Engineering

[Statistical Modeling](#) · [Time Series Analysis](#) · [Python](#) · [Health Data](#) · [EEG](#)

- Analyzing high-dimensional EEG time-series data to develop predictive models for mental health conditions — directly relevant to clinical AI and medical data processing at scale.
- Applying advanced statistical techniques to extract actionable insights from complex biometric datasets — the same class of problem as structuring and analyzing medical records for model training.
- Collaborating with domain experts across engineering and health sciences, bridging the gap between technical implementation and clinical relevance.

AI-Powered Mental Health Support Platform

Aug 2025

HKU · [github.com/jintaoxie02/mental-health-counselling-bot](#)

[Python](#) · [LangChain](#) · [RAG](#) · [Next.js](#) · [TypeScript](#) · [Data Privacy](#)

- Built an AI-driven counseling platform with LangChain conversation pipeline and RAG patterns — demonstrating production AI application in a healthcare context.
- Designed confidentiality-first data architecture and messaging flows — aware of the privacy and compliance standards required in health tech (HIPAA-adjacent thinking).

Memory-OS — Hierarchical Memory System for AI Agents

Jul 2026

[github.com/jintaoxie02/memory-os](#) · MIT License · 37 tests

[Python](#) · [LLM Integration](#) · [LangChain](#) · [ChromaDB](#) · [SQLite](#) · [Agent Architecture](#)

- Designed and built a three-layer memory hierarchy for LLM agents — clean, well-tested library architecture (37 tests) demonstrating strong software engineering and API design skills.
- Implemented extraction, merging, and retrieval engines with hybrid scoring — the kind of modular, scalable AI infrastructure relevant to building medical AI systems.

InsurPrime: Insurance Premium Prediction — Top 11% Globally

Jan 2026

Challenge Data / ENS Paris — Organized by Crédit Agricole Assurances

[Python](#) · [Machine Learning](#) · [Predictive Modeling](#) · [Insurance Analytics](#)

- Ranked 42nd (top 11.05%) globally predicting insurance premiums from real-world policyholder data — demonstrates applied ML on structured health-relevant datasets.

Production ML — Probability & Risk Modeling Pipeline

Jun 2026

[github.com/jintaoxie02/hkjc-football-model](#) · [github.com/jintaoxie02/HKJC_Horse_v2](#)

[Python](#) · [XGBoost](#) · [LightGBM](#) · [CatBoost](#) · [DuckDB](#) · [Flask](#) · [Chrome Extension](#)

- End-to-end ML pipeline: walk-forward validated across 16,886 samples, 113 leagues, Sharpe ~1.56, profitable in all 10 folds. Full-stack deployment with live server (Flask) and Chrome extension — demonstrates ability to build, deploy, and ship complete ML systems.

EXPERIENCE

Programmer (Freelance) — AIGreen

Jul - Sep 2025

Hong Kong SAR · Hybrid

- Developed control software with hardware interfacing (Arduino, ESP32) — rapid prototyping, iterative testing, and independent delivery in a fast-paced environment similar to startup cadence.
- Delivered frequent updates and coordinated requirements directly with stakeholders.

EDUCATION & AWARDS

BEng Data Science & Engineering, Minor in Digital Health

2023 – 2027

The University of Hong Kong

HKU Engineering Recognition Ceremony 2024–25 · Winner, HKAE Pitch Competition 2024–2025

Relevant coursework: Digital Health, Statistical Modeling, Machine Learning, Data Privacy

LANGUAGES

Cantonese (Native) · Mandarin (Native) · English (Fluent)

Available immediately · Part-time · Flexible hours · [github.com/jintaoxie02](#)